

57. (d) A number is divisible by 16 if the number formed by its last 4 digits is divisible by 16.

∴ Required remainder

$$= \text{Rem.} \left(\frac{1920}{16} \right) = 0$$

58. (b) Required probability = $\frac{{}^4C_2 \times {}^8C_1}{{}^{12}C_3}$

$$= \frac{6 \times 8}{220} = \frac{12}{55}$$

59. (c) Required number of carpets

$$= \frac{960}{6 \times 4} = 40$$

60. (b) Let the distance by d km.

$$\text{Then, } \frac{d}{3} - \frac{d}{4} = \frac{15+15}{60}$$

$$\Rightarrow \frac{d}{12} = \frac{1}{2} \Rightarrow d = 6$$

61. (a) Area left for writing = $(30 - 2 \times 2.5) \times (15 - 2 \times 1.5) = 25 \times 12 \text{ cm}^2$

Required percentage

$$= \frac{25 \times 12}{30 \times 15} \times 100 = 66.66\%$$

62. (b) Let the weight of Ram and Shyam be 6 kg and 5 kg respectively.

$$\text{Increased weight of Shyam} = (6 + 5) \times 1.15 - 6 \times 1.2$$

$$= 12.65 - 7.2 = 5.45 \text{ kg}$$

$$\text{Required percentage} = \frac{5.45 - 5}{5} \times 100 = 9\%$$

63. (a) Total quantity of water in new mixture

$$= 2 + \frac{1}{7} \times 14 = 4 \text{ litres}$$

Total quantity of milk in new mixture

$$= (8 + 14) - 4 = 18 \text{ litres}$$

Required ratio = 4 : 18 = 2 : 9

64. (d) $P \rightarrow CP = x$, Profit = $0.4x$, SP = $1.4x$

$$Q \rightarrow SP = 1.4x$$

$$\text{Profit} = 0.5 \times 1.4x = 0.7x$$

$$\text{Now, } 0.7x - 0.4x = 600$$

$$\Rightarrow 0.3x = 600 \Rightarrow x = 2000$$

$$\therefore SP = 1.4 \times 2000 = 2800$$

65. (b) Let the age of the youngest child be x years.

According to the question,

$$25 \times 5 + 8 \times 5 + x + x + 4 = 25 \times 7$$

$$\Rightarrow x + x = 175 - 169$$

$$\Rightarrow 2x = 6 \Rightarrow x = 3$$

66. (a) Distance travelled by the truck in 10 litres of diesel at the speed of 50 km/h

$$= 10 \times 19.5 = 195 \text{ km}$$

$$\therefore \text{Required distance} = \frac{195}{1.3} = 150 \text{ km}$$

67. (a) At 't' minutes after 2:00,

$$\text{Hour hand angle} = 60 + 0.5t \text{ degrees}$$

$$\text{Minute hand angle} = 6t \text{ degrees}$$

$$\text{As, 1 minute space} = 6 \text{ degrees,}$$

$$\text{So, } 6t - (60 + 0.5t) = 6$$

$$\Rightarrow 5.5t = 66 \Rightarrow t = 12$$

Hence, at 2:12 the minute hand is exactly 1 minute space ahead of hour hand.

C1 C2 C3

68. (c) Profit $\rightarrow 9 : 10 : 8$

$$81 : 90 : 72$$

C2 C4 C5

$$\text{Profit} \rightarrow 18 : 19 : 20$$

$$90 : 95 : 100$$

∴ Profit ratio of C1, C2, C3, C4 and C5

$$= 81 : 90 : 72 : 95 : 100$$

$$\therefore \text{Total profit} = \frac{19}{100 - 81} \times (81 + 90 + 72$$

$$+ 95 + 100) = ₹ 438 \text{ Cr}$$

69. (a) Let the age of younger child be x years.

According to question,

$$4 \times 40 + 4 \times 15 + x + x + 8 = 6 \times 40$$

$$\Rightarrow 228 + 2x = 240 \Rightarrow x = \frac{12}{2} = 6$$

$$\text{Required ratio} = (6 + 8) : 6 = 7 : 3$$

70. (b)

Ramesh

Rakesh

$$\text{CP} \rightarrow x$$

$$\text{Profit} \rightarrow \frac{x}{4}$$

$$\frac{1}{4} \times \frac{5}{4} x = \frac{5}{16} x$$

$$\text{SP} \rightarrow x + \frac{x}{4} = \frac{5}{4} x$$

$$\frac{5}{4} x$$

Now,

$$\frac{5x}{16} - \frac{x}{4} = 100 \Rightarrow \frac{x}{16} = 100 \Rightarrow x = 1600$$

$$\text{SP} = \frac{5}{4} \times 1600 = ₹ 2000$$

71. (c) $\frac{2}{3}$ day $\rightarrow 18$ books

$$1 \text{ days} \rightarrow 18 \times \frac{3}{2} = 27 \text{ books}$$

$$\frac{1}{4} \text{ day} \rightarrow \frac{27}{4} \text{ books}$$

72. (b) Upstream speed = $\frac{140}{7} = 20 \text{ km/h}$

$$\text{Downstream speed} = \frac{210}{7} = 30 \text{ km/h}$$

∴ Speed in still water

$$= \frac{20 + 30}{2} = 25 \text{ km/h}$$

73. (d) Perimeter of rectangle = Circumference of circle

$$= 2 \times \frac{22}{7} \times 21 = 132 \text{ cm}$$

$$\therefore 2(6x + 5x) = 132 \Rightarrow 22x = 132 \Rightarrow x = 6$$

$$\text{Area of rectangle} = 6x \times 5x = 30x^2$$

$$= 30 \times 6^2 = 1080 \text{ cm}^2$$

74. (c) $\sqrt{1225} \div \sqrt[3]{343} \times 45$ % of 760

$$= x \times 66.67\% \text{ of } 45$$

$$\Rightarrow 35 \div 7 \times \frac{45}{100} \times 760 = x \times \frac{2}{3} \times 45$$

$$\Rightarrow 5 \times 342 = x \times 30 \Rightarrow x = \frac{342}{6} = 57$$

75. (a)

A B

$$\text{Time} \rightarrow 20 : 30$$

$$2 : 3$$

$$\text{Efficiency} \rightarrow 3 : 2$$

Required difference

$$= \frac{3-2}{3+2} \times 4000 = ₹ 800$$

76. (b) 1 minute-space = 6° .

At 't' minutes past 3:00

$$\text{Minute hand angle} = 6t^\circ$$

$$\text{Hour hand angle} = 90^\circ + 0.5t^\circ \text{ (since at 3:00 it's at } 90^\circ \text{ and moves } 0.5^\circ/\text{min)}$$

Minute hand ahead by 7 spaces

$$\Rightarrow \text{minute} - \text{hour} = 7 \times 6 = 42^\circ$$

$$\Rightarrow 6t - (90 + 0.5t) = 42$$

$$\Rightarrow 5.5t - 90 = 42$$

$$\Rightarrow 5.5t = 132$$

$$\Rightarrow t = 24 \text{ minutes}$$

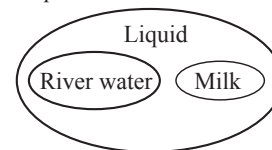
So, the time is 3:24.

77. (a) Logic: They're the two-letter suffixes we add to numbers when writing ordinal numbers in English.

As, 1 $\rightarrow$ st (1st), 2 $\rightarrow$ nd (2nd), 3 $\rightarrow$ rd (3rd), 4 $\rightarrow$ th (4th)

Similarly, Continuing the sequence (1st, 2nd, 3rd, 4th), the next is 5th, whose suffix is TH.

78. (a) Logic: "River water" and "Milk" are Liquid.

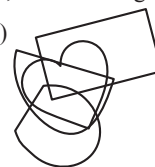


79. (d) After reversing each half:

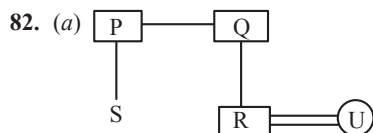
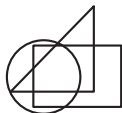
MLKJIHGFEDCBZYXWVUTSRQPON

So, 10th from right is W.

80. (c)



81. (c)



Here, U is the Sister in law of S.

83. (b) Logic: In Numerator each letter is decreased by 1 and denominator decreased by 3.

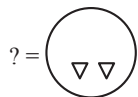
As,

F N B and V
 $\downarrow -1$ $\downarrow -1$ $\downarrow -1$ $\downarrow -3$
 E M A S

Similarly,

G P C and Z
 $\downarrow -1$ $\downarrow -1$ $\downarrow -1$ $\downarrow -3$
 F O B W

84. (a) Logic : Second image is the water image of the first image in each pair.



85. (b) Political Science, History and Sociology are the parts of social sciences. But, Biology is natural science.

86. (c) Except number 15, all other numbers are even number.

87. (a) Logic : First reverse the word and then subtract 3, 2 alternatively.

As,

MOBILE in reverse order ELIBOM,

then,

E L I B O M
 $\downarrow -3$ $\downarrow -2$ $\downarrow -3$ $\downarrow -2$ $\downarrow -3$ $\downarrow -2$
 B J F Z L K

and,

TABLET in reverse order TELBAT,

then,

T E L B A T
 $\downarrow -3$ $\downarrow -2$ $\downarrow -3$ $\downarrow -2$ $\downarrow -3$ $\downarrow -2$
 Q C I Z X R

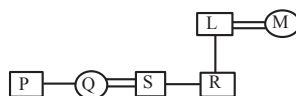
Similarly,

KINDLE in reverse order ELDNIK,

then,

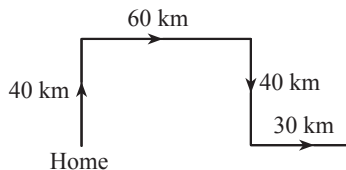
E L D N I K
 $\downarrow -3$ $\downarrow -2$ $\downarrow -3$ $\downarrow -2$ $\downarrow -3$ $\downarrow -2$
 B J A L F I

88. (b)



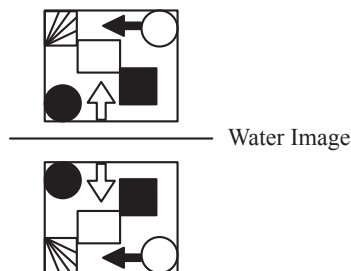
Here, P is S's wife's brother.

89. (b)

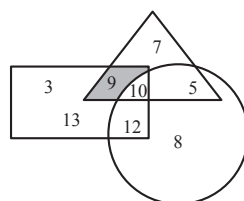


Shilpa is 90 km far from her home.

90. (d)

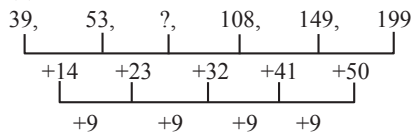


91. (a)



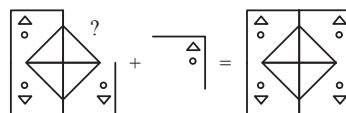
Here, 9 players play both kabaddi and football only.

92. (d)



$$? = 53 + 23 = 76$$

93. (a)



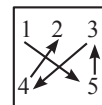
94. (b) Given : $105 - 15 \div 13 + 3 \times 27 = ?$

After interchanging '+' with ' $\times$ '; and '-' with ' $\div$ ',

$$= 105 \div 15 - 13 \times 3 + 27$$

$$= 7 - 39 + 27 = -5$$

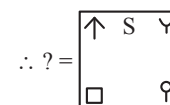
95. (a)



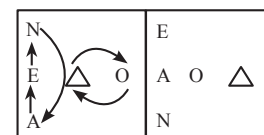
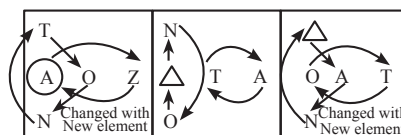
Logic :-

I. Letters or symbols changes its position in $1 \rightarrow 5, 5 \rightarrow 3, 3 \rightarrow 4$, and $4 \rightarrow 2$ sequence.

II. In the next series new symbols comes in position 1.



96. (b)

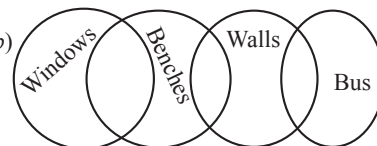


97. (a)



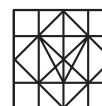
Ranjita and Fauzia sitting at the extreme ends.

98. (b)



Only Conclusion II follows.

99. (b)



100. (c)

